A Distributed Combinatorial Topology Approach to Arrow's Impossibility Theorem

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ABSTRACT

Baryshnikov presented a remarkable algebraic topology proof of Arrow's impossibility theorem trying to understand the underlying reason behind the numerous proofs of this fundamental result of social choice theory. We present here a novel combinatorial topology approach that does not use advanced mathematics, while giving a geometric intuition of the impossibility. This exposes a remarkable connection with distributed computing techniques.

We study in detail the case of two voters, $n = 2$, and three alternatives, $|X| = 3$, and show that Arrow's impossibility is closely related to the index lemma. Also, we study the domain restrictions that avoid the impossibility. Finally, we explain why the case of $n = 2$ and $|X| = 3$ is where this interesting geometry happens, by giving a simple proof that this case implies Arrow's impossibility for any $|X| \geq 3$ and any finite $n \geq 2$.

CCS CONCEPTS

• **Mathematics of computing** → **Algebraic topology**; **Combinatorics**.

KEYWORDS

Combinatorial Topology, Arrow's impossibility theorem, index lemma, domain restriction, social choice

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1 INTRODUCTION

The modern field of social choice theory took off with Kenneth Arrow's remarkable 1950 result [3] for the basic problem of democracy: it is impossible to aggregate individual preferences into a single social preference, under some reasonable-looking axioms[1]. Consider a set of $n \geq 2$ voters and $|X| \geq 3$ alternatives. Each voter ranks the alternatives in a total order. An aggregation function F takes as input the n total orders, and produces a single total order of the alternatives, the social preference. The reasonable-looking axioms are the following three. Unanimity: if alternative, a is ranked higher than b by all voters, then so it is in the social preference. Non-dictatorship: There is no voter k whose preferences always prevail. Independence of irrelevant alternatives: the social preference between a and b is not affected by individual preferences between other alternatives. A formal definition of Arrow's theorem is in Section 2.1.

Soon after the publication of Arrow's result alternative proofs began to emerge; starting with Inada [36] in 1954, numerous other proofs followed, and continue to be proposed until recently, e.g. [25, 32]. Its publication also triggered an extensive discussion about various other issues, such as alternative models for aggregating preferences (e.g. probabilistic [9], fuzzy [22], etc), and about whether Arrow's axioms are too restrictive [14]. We are interested in another issue that has also received much attention: the study of restrictions of the individual preferences, where non-dictatorial aggregation is possible [29]. To the best of our knowledge, a full characterization of such domains restrictions remains open.

Motivation. Trying to understand the underlying reason behind the many proofs of Arrow's theorem, Baryshnikov [10] presented in 1993 a striking different approach, a topological impossibility proof. He discussed in [11] the value of the approach: *"the topology we are exploiting is in fact very geometrical and thus appeals to our intuition much better than dry combinatorial constructions"*. However, the goal of providing intuition about the nature of the problem of social choice is hindered by the relatively advanced algebraic topology tools used by Baryshnikov (several attempts at explaining the proof have been made [11, 19, 47]).

Our goal here is to further advance the program of Baryshnikov, while making it accessible to an audience not familiar with algebraic topology. Furthermore, we aim at closing the gap between Baryshnikov's proof and combinatorial proofs, which have remained largely independent. We do so by moving from algebraic topology to combinatorial topology, and in doing so discover (and benefit from) remarkable connections with distributed computing.

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[1] Social choice theory is a highly developed field, of interest to economics and political science. For an overview, including the importance of Arrow's result, see introductory books such as [30], or more advanced such as [24]. It is also of interest to topology [47], and more recently to computer science [13].

Contributions. First we concentrate on the case of two voters, $n = 2$, and three alternatives, $|X| = 3$. We provide a new proof of Arrow's impossibility that does not require any acquaintance with algebraic topology. The proof gives a new insight for the reason of the impossibility: the index lemma, an important combinatorial topology result used to compute winding numbers.

We then show the usefulness of the combinatorial topology approach to study domain restrictions. We prove that contractibility of the space of restricted preference profiles is not what determines the existence of non-dictatorial aggregation; the arguments based on a domain restriction explain the index lemma impossibility argument in a very intuitive geometric way. Furthermore, the arguments lead to a full characterization of the domain restrictions where non-dictatorial aggregation is possible.

Finally, we aim at explaining an intriguing phenomenon, appearing several times in the literature: the case of $n = 2$, and $|X| = 3$ is somehow special. Some papers simply treat this case only e.g. [2, 21, 47, 51]. More interestingly, Baryshnikov [10, Section 7.1] hints at the idea that this is the case where the interesting things happen. He explains that only the arguments of his proof for triples of alternatives are in fact used, and one could concentrate only on the 2-skeleton of the simplicial complex using one-dimensional (co)homology. Our third result is a simple proof showing that Arrow's impossibility result for the case of $n = 2$ and $|X| = 3$ implies the general case. This result has been shown before under the restriction of finite $|X|$ by Tang and Lin [52] and partially by Akashhi [2], but our proof seems more direct, in addition to being more general.

For lack of space, we defer some proofs and additional results to the full version [49]. In particular, we present a combinatorial topology perspective of the recent pivotal arguments to prove Arrow's impossibility by Geanakoplos [32] and Yu [56] that have received much attention e.g. [55]. Notice that Baryshnikov [10] does not try to explain the relation of his topology proof with previous proofs.

Organization. In Section 2 we present the statement of Arrow's theorem, and how to model it using combinatorial topology. In Section 3 we provide a proof of Arrow's combinatorial topology theorem, when $n = 2, |X| = 3$, using the index lemma. In Section 4 we present the results related to domain restrictions. In Section 5 we present the argument to generalize Arrow's theorem from the basis case of $n = 2, |X| = 3$. In Section 6 we discuss relations to distributed computing, and in Section 7 we present the conclusions.

2 ARROW'S IMPOSSIBILITY THEOREM

We start by recalling Arrow's theorem in Section 2.1, we then give an informal presentation of the approach in Section 2.2, and finally we translate it into a combinatorial topology theorem in Section 2.3.

2.1 Problem statement

Let X be a set of alternatives, $|X| \geq 3$. The set of all strict total orders of X is denoted by W. Let $n \geq 2$ denote the (finite) number of voters, and W^n be the set of profiles of preferences. Thus, $\mathbf{R} = (R_1, \ldots, R_n) \in W^n$ is a profile, where each R_i is the order on X preferred by the i-th voter, $R_i \in W$. An aggregation map F is a function from W^n to W that maps each profile of W^n to a unique order in W. For example, if $X = \{A, B, C\}$, $R_i = A \succ B \succ C \in W$ denotes that the i-th voter prefers A over B, and B over C. This is also denoted as $A\, R_i\, B\, R_i\, C$, or when no confusion arises, simply by ABC. A classic form of Arrow's impossibility is the following.

Theorem 2.1 (Arrow's impossibility). *Assume $|X| \geq 3$. There is no aggregation map F from W^n to W satisfying the following axioms.*

1. *Unanimity. If alternative, a, is ranked strictly higher than b for all orderings $R_1, \ldots, R_n$, then a is ranked strictly higher than b by $F(R_1, \ldots, R_n)$.*
2. *Non-dictatorship. There is no $k \in \{1, \ldots, n\}$ such that for all $\mathbf{R} \in W^n$, a ranked strictly higher than b by R_k implies a ranked strictly higher than b by $F(R_1, \ldots, R_n)$, for all a and b.*
3. *Independence of irrelevant alternatives. For two preference profiles $\mathbf{R}$ and $\mathbf{S}$ such that for all individuals i, alternatives a and b have the same order in R_i as in S_i, alternatives a and b have the same order in $F(R_1, \ldots, R_n)$ as in $F(S_1, \ldots, S_n)$.*

Some formulations of Arrow's impossibility theorem allow ties in the rankings (e.g. [4, 27, 57]). In this sense, it could seem that the framework we present here is not as general as it might be. However, this is not the case e.g. [10, Lemma 1], and indeed previous proofs e.g. [10, 39, 47] of Arrow's impossibility often assume, as we do, strict orders.

2.2 Informal overview of the combinatorial topology form of Arrow's theorem

Overview. The intuition behind our approach, for the base case of two voters and three alternatives A ,B and C is the following. The first step is to represent the set of possible preferences of the voters, N_I, as well as the set of possible social preferences, N_O, as geometric objects built from triangles. We illustrate the set of triangles of N_I and N_O in Figure 1. Each triangle of N_I represents a social profile, and it is mapped by a simplicial map f induced by the aggregation map F to a triangle of N_O representing the corresponding social choice.

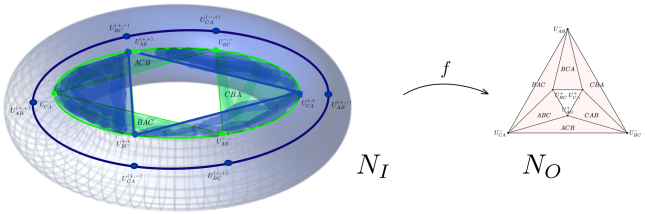


Figure 1: On the left, N_I is a torus with 12 additional triangles, 6 attached to the green cycle, and 6 attached to the blue cycle (not shown). Each such triangle shares two edges with the corresponding cycle; the other edge is part of the boundary of N_I. The boundary egdes form four boundary hollow triangles. Notice that N_O is homeomorphic to a cylinder; it has two hollow boundary triangles.

The notation N_I, N_O stands for "input" and "output" complexes of distributed computing (see Section 6).

The remarkable insight is to observe that f is a simplicial map: it sends triangles of N_I to triangles of N_O, and if two triangles share a vertex (edge) in N_I then f must send them to two triangles in N_O that also share a vertex (edge). This follows from the independence

of irrelevant alternatives axiom. In Section 2.3 we present a combinatorial topology reformulation of Arrow's theorem, stating that any simplicial map f satisfying unanimity must be a dictatorship.

The complexes. We use the simplicial complexes N_I and N_O on which Baryshnikov's framework [10] is based. These simplicial complexes have been explained in detail e.g. [19, 40, 47] where additional details can be found.

The intuition behind these complexes is as follows, for $|X| = 3, n = 2$. On the right side of Figure 2, two triangles of N_O are depicted, labeled *BAC* and *BCA*. The label *BAC* means that society prefers B over A and A over C. The two triangles share an edge because *BAC* and *BCA* agree on two pairwise preferences. The first is represented by the vertex U^-_{AB}, namely B is preferred over A, and the second by the vertex U^+_{BC}, namely B is preferred over C. The

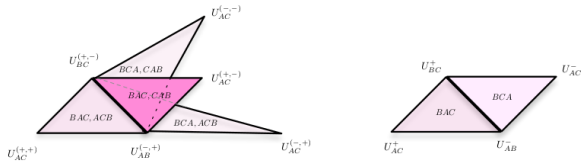


Figure 2: Four triangles of N_I intersecting in an edge, and two of N_O intersecting in an edge.

triangles of N_I are similarly defined, by using the preferences of two voters. For instance, in the the triangle labeled *BAC*, *ACB*, the first voter prefers *BAC* while the second voter prefers *ACB*. On the left part of Figure 2, an edge is contained in four triangles, representing four different profiles, all sharing their pairwise preferences for A, B and B, C. In all 4 triangles, the first voter prefers *BA* and *BC*, while the second prefers *AB* and *CB*.

2.3 Combinatorial topology form of Arrow's theorem

The main goal of this section is to state Arrow's theorem in the combinatorial topology framework.

We define the two simplicial complexes, N_I, N_O, for a finite set of alternatives, $|X| \geq 3$ and $n \geq 2$ voters. The first one, N_I represents all possible profiles of preferences, and the second, N_O, represents all possible social preferences. A simplicial map f from N_I to N_O represents an aggregation map that satisfies the independence of irrelevant alternatives property. As we shall see, if f sends vertices where everybody prefers α over β, denoted $U^{(+,\cdots,+)}_{\alpha\beta}$, to vertices where α is preferred over β in the social choice, denoted $U^+_{\alpha\beta}$, then f also satisfies the unanimity property. Finally, there is a dictator if f is a projection. Intuitively, the following says that Arrow's theorem can be viewed as the existence of a continuous map from N_I to N_O preserving unanimity. We will explain in detail the theorem in Sections 2.3.1, 2.3.2 and 2.3.3.

Theorem 2.2 (Arrow's impossibility). *Let $|X| \geq 3$ and $n \geq 2$. If $f : N_I \to N_O$ is a simplicial map such that for all vertices $U^{(+,\cdots,+)}_{\alpha\beta}$ of N_I, it holds that $f(U^{(+,\cdots,+)}_{\alpha\beta}) = U^+_{\alpha\beta}$, then f is a projection.*

That f is a projection means that there is a dictator k, such that, f returns the preferences of the k-th voter. That is, as we explain in more detail below, for all vertices $U^{\boldsymbol{\sigma}}_{\alpha\beta}$ of N_I,

$$f(U^{\boldsymbol{\sigma}}_{\alpha\beta}) = U^{\boldsymbol{\sigma}(k)}_{\alpha\beta},$$

where $\boldsymbol{\sigma}(k) \in \{+,-\}$ is the k-th sign of the vector of n signs $\boldsymbol{\sigma}$.

2.3.1 The output complex N_O. We define the *output complex* N_O as follows. Consider the notation $U^{\sigma}_{\alpha\beta}$, for $\alpha, \beta \in X$ with $\alpha \neq \beta$ and $\sigma \in \{+,-\}$. Then, $U^{\sigma}_{\alpha\beta}$ denotes the subset of W, of all strict orderings on X such that α is ranked higher than β if $\sigma = +$, and otherwise, β is ranked higher than α. Notice that $U^+_{\alpha\beta}$ denotes the same set as $U^-_{\beta\alpha}$. The set vertices V of the output complex N_O consists of all such subsets of W, each one identified by one $U^{\sigma}_{\alpha\beta}$. A set of vertices of V forms a simplex of N_O iff their intersection is nonempty. This family of sets forms a simplicial complex, as it is closed under containment.

As mentioned earlier, for the purposes of this article, it is sufficient to consider $|X| = 3$. Then, the complex N_O is depicted in Figure 1 taking $X = \{A, B, C\}$. We remark that our discussion holds for any finite X.

In the case of $|X| = 3$, N_O is of dimension 2. A facet is a 2-simplex $\{U^{\sigma_0}_{\alpha_0\beta_0}, U^{\sigma_1}_{\alpha_1\beta_1}, U^{\sigma_2}_{\alpha_2\beta_2}\}$, which represents the strict order that is compatible with its three vertices, that is, the strict order contained in $U^{\sigma_0}_{\alpha_0\beta_0} \cap U^{\sigma_1}_{\alpha_1\beta_1} \cap U^{\sigma_2}_{\alpha_2\beta_2}$. Consider for example the triangle ABC, and its two vertices U^+_{AB} and U^+_{BC}. Notice that $U^+_{AB} = \{ABC, ACB, CAB\}$, and $U^+_{BC} = \{ABC, BAC, BCA\}$. These two vertices form an edge of N_O because their intersection is not empty. Moreover, it belongs to a single triangle, because the third vertex is unique, $U^-_{CA} = \{ABC, ACB, BAC\}$. Indeed, the three vertices intersect in a unique order, ABC.

There are exactly two triangles that are empty, that do not form a simplex, the external one requiring that $A \succ B, B \succ C, C \succ A$, and the central one, requiring that $A \succ C, C \succ B, B \succ A$. Furthermore, the boundary edges that belong to a single triangle are those that by transitivity uniquely imply the third vertex, e.g. the edge $\{U^+_{AB}, U^+_{BC}\}$ implies the third vertex, U^-_{CA}. Similarly, a partial order defined by an edge, e.g. $\{U^+_{AB}, U^+_{AC}\}$, is compatible with the two vertices that resolve the incomparability of B and C, namely, U^-_{BC} and U^+_{BC}.

The complex N_O is the space of output preferences because each one of its facets represents a possible social preference. Such a social preference is decided by an aggregation rule f, applied to a set of individual preferences of W^n, which we also represent as a simplicial complex.

Remark 2.1. For simplicity, we always denote the six vertices of N_O by the representatives $U^+_{AB}, U^-_{AB}, U^+_{BC}, U^-_{BC}, U^+_{CA}$ and U^-_{CA}, as in the figure. In Section 3 we will need all vertices in the same boundary to share the same sign.

Remark 2.2. Consider two *adjacent* 2-simplices, intersecting in an edge. The strict order associated with one simplex and the one associated with the other simplex are equal, modulo permuting two consecutive elements in the strict order. For example, the facet corresponding to *ABC* and the one corresponding to *ACB* are adjacent: they are equal modulo the permutation of B and C.

2.3.2 The input complex N_I. We define the sets $U^{\boldsymbol{\sigma}}_{\alpha\beta}$, with $\alpha, \beta \in X$ and $\boldsymbol{\sigma} \in \{+,-\}^n$ as the subset of profiles of W^n where for each voter i, α is ranked higher than β if $\boldsymbol{\sigma}(i) = +$, and otherwise, β is ranked higher than α. As before, $U^{\boldsymbol{\sigma}}_{\alpha\beta}$ defines the same set of social preferences as $U^{-\boldsymbol{\sigma}}_{\beta\alpha}$. The set of vertices of the input complex N_I consists of all such subsets of W^n. As in the previous section, a set of vertices is a simplex of N_I iff their intersection is nonempty.

The complex N_I is much bigger than N_O. Whereas N_O has $|X|(|X|-1)$ vertices and its dimension is $(|X|+1)(|X|-2)/2$, N_I has $|X|(|X|-1)2^{n-1}$ vertices, but it has the same dimension as N_O (see [10]). So, in contrast to N_O, the complex N_I cannot be drawn in the plane even when $|X| = 3$, but a schematic representation is in Figure 1 and in Figure 3.

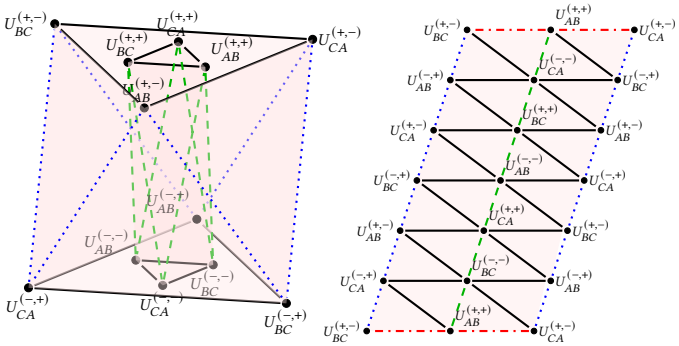


Figure 3: When $|X| = 3$ and $n = 2$, the complex N_I can be built using two (cylindrical) copies of N_O placed one inside the other (on the left side of the figure). The outer cylinder are the unanimous profiles, whereas the inner one are the profiles where the voters have opposite preferences. Additionally, both cylinders are joined through the torus in the right (the torus is folded by identifying vertices according to the coloured edges), so the total number of vertices of N_I is 12.

Remark 2.3. Consider two adjacent 2-simplices (intersecting in an edge) of N_I, $|X| = 3$, $n = 2$. The individual preferences associated with one simplex and those associated with the other simplex are equal, modulo permuting the preference of two alternatives, x, y, of one or two voters, without changing the preferences of other alternatives. For example, in Figure 2, the triangles BAC, ACB and BCA, CAB are adjacent, because the preferences of both voters over A and C are exchanged, and only over A and C.

As an example, consider the inner cylinder on the left of the Figure 3. The front triangle has vertices $U^{(-,-)}_{CA}, U^{(+,+)}_{AB}, U^{(+,+)}_{BC}$. This represents that both voters prefer ABC. The vertex $U^{(+,+)}_{AB}$ is also contained in its right triangle where both prefer CAB. The green edge of this triangle, $\{U^{(+,+)}_{AB}, U^{(-,-)}_{BC}\}$, is contained in the triangle (in the torus on the right side of the figure) that also contains $U^{(+,-)}_{CA}$, representing that the first voter prefers CAB but the second prefers ACB. Figure 1 illustrates how N_I consists of a torus, where two "parallel" cycles, a green one and a blue one are identified with some additional triangles (in the figure only the triangles identified with the green one are drawn, for clarity). In the green cycle, 6 vertices are used to add 6 triangles, as "flaps" of the torus (same for the blue cycle).

2.3.3 The simplicial aggregation map. Finally, it remains to define an aggregation simplicial map $f : N_I \to N_O$ from an aggregation map $F : W^n \to W$. Since F satisfies the independence of irrelevant alternatives property and $U^{\boldsymbol{\sigma}}_{\alpha\beta}$ is a subset of profiles in W^n defined purely by the orderings between α and β, $f(U^{\boldsymbol{\sigma}}_{\alpha\beta})$ can be defined to be the vertex $U^{\sigma}_{\alpha\beta}$ with the sign σ determined by the ordering of α and β on the social aggregation of any of the profiles in $U^{\boldsymbol{\sigma}}_{\alpha\beta}$.

The images of the higher dimensional simplices of N_I can be defined by extension. We only need such simplices to be in N_O. However, this is immediate because a simplex in N_I is generated iff the intersection of their vertices contains at least one profile. The image of such a profile must belong to the intersection of the images of those vertices since the image of a profile is determined by the ordering of pairs of alternatives.

Finally, we get the statement of Theorem 2.2. The independence of irrelevant alternatives property implies that f is a simplicial map from N_I to N_O. Moreover, the unanimity of f determines the image of the vertices formulated as $U^{(+,\cdots,+)}_{\alpha\beta}$ or $U^{(-,\cdots,-)}_{\alpha\beta}$.

3 IMPOSSIBILITY PROOF BASED ON THE INDEX LEMMA

We present the proof of Theorem 2.2, for $|X| = 3, n = 2$, using a simple generalization of the index lemma [34].

3.1 Intuition

Cycles (empty triangles) defined by *boundary* edges turn out to be important. An edge in the boundary of a complex is contained in a single triangle because the pair of vertices of the edge determines the third one by transitivity. Consider for example the two pairwise preferences of ABC given by the vertices U^{+}_{AB} and U^{+}_{BC} in Figure 1. This edge belongs only to this triangle, since the vertices together determine the order ABC.

Consider for example the edge representing that both voters prefer AB and both prefer BC, given by the vertices $U^{(+,+)}_{AB}$ and $U^{(+,+)}_{BC}$ (see Figure 1). This is a boundary edge because it is contained in the unique triangle where both prefer ABC. Three such edges (connecting the two former vertices with $U^{(+,+)}_{CA}$) form a hollow triangle, because a Condorcet cycle is created if also both of them prefer CA. There are edges in N_I that are internal, for instance when both prefer AB, while the first voter prefers BC and the second prefers CB, see Figure 6b. This edge is contained in two triangles; in the figure, the left triangle correspond to the first voter's preference ABC, and the second voter's preference ACB. The other triangle in the figure corresponds to the preferences ABC and CAB. There are also internal edges contained in four triangles, as illustrated in Figure 2.

The aggregation map f maps (hollow) boundary triangles to (hollow) boundary triangles, as illustrated in Figure 1. Notice that

N_O is a triangulation of a cylinder with two boundary triangles, while N_I is a kind of product of two cylinders and has 4 boundary triangles. The index lemma computes the winding number[2] of the boundary triangles of N_I on the boundary triangles of N_O. As we shall see, this number is 0 and implies that f is a projection (on the preferences of one of the two voters, the dictator), assuming that f satisfies unanimity. The mathematics used is elementary: essentially only basic parity counting operations are needed.

3.2 The index lemma

Let K be a simplicial complex of dimension 2 satisfying that every simplex of dimension 1 is contained in a single or in an even number of 2-simplices. An *orientation on* K is an orientation on every 2-simplex satisfying that the induced orientations on the 1-simplices by the 2-simplices have to be opposite by pairs.

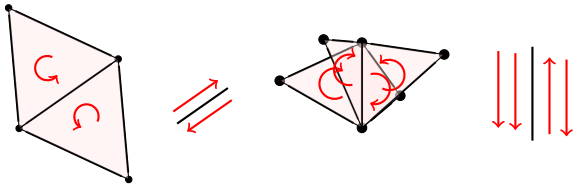


Figure 4: The simplicial complex on the left is oriented because the induced orientations on the inner edge are opposite. However, the right one is not because it has three orientations in one direction and one on the opposite direction.

Let K be an oriented simplicial complex of dimension 2 with each vertex labeled with a color from $\{0, 1, 2\}$. The *content* C of K is the number of tricoloured triangles in K counted +1 if the order of the labeling agrees with the orientation (see the right side of Figure 5) and -1 otherwise. The *index* I of K is the number of edges $\overrightarrow{01}$ on the boundary counted +1 if the order of the vertices agrees with the orientation and -1 otherwise. The original version of the index lemma is for the case that every simplex of dimension 1 is contained in a single or in two 2-simplices, here we prove the generalized version, for the case that a 1-simplex may be contained in an even number of 2-simplexes.

Theorem 3.1 (Index lemma). *If K is a 3-colored oriented simplicial complex of dimension 2, then $I = C$.*

Proof. Let S be the number of edges $\overrightarrow{01}$ counted according to the orientation. We will prove that $I = S$ and $C = S$. First, we will see that the contribution of every interior edge $\overrightarrow{01}$ is equal to 0. Since every interior edge has an even number of incident 2-simplices, by definition of being oriented, their contribution is 0. Then $I = S$.

For every triangle in the complex, the contribution is only non-zero if the triangle is tricoloured. If it is not tricoloured, it is 0 because, in case it has at least one 0 and one 1, the third vertex has to be coloured by 0 or 1, then one edge compensates the other. Otherwise, if it is tricoloured, its contribution is the same as the content's contribution (see Figure 5). □

[2]The *winding number* of a closed curve in the plane around a point is the number of times that the curve passes counterclockwise around the point minus the number of times it passes clockwise. It is important in topology, calculus, analysis, physics, etc. The index lemma is a generalization of Sperner's lemma, which is equivalent to Brouwer's fixed point theorem.

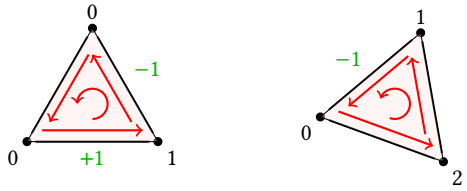


Figure 5: On the left, the contribution of the simplex is 0 because the two edges $\overrightarrow{01}$ compensate each other. On the right, the contribution of the tricolored triangle is -1.

3.3 The proof based on the index lemma

First we provide N_I with an orientation.

Lemma 3.2. *The complex N_I is orientable.*

Proof. We will define an orientation on N_I as follows. For every 2-simplex $\Delta = \{U_{AB}^{\sigma_1}, U_{BC}^{\sigma_2}, U_{CA}^{\sigma_3}\}$ we define its parity as the product of all the signs of σ_1, σ_2 and σ_3. For instance, if $\sigma_1 = (+,+)$, $\sigma_2 = (+,-)$ and $\sigma_3 = (-,-)$, the parity is -1 (see Figure 6a). We define the orientation of this 2-simplex as clockwise ($AB \to CA \to BC \to AB$) if its parity is -1 and ($AB \to BC \to CA \to AB$) if its parity is 1.

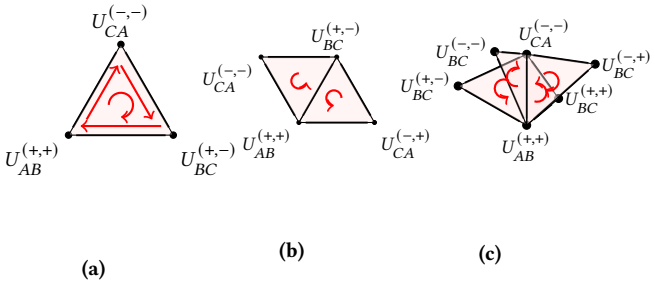


Figure 6: (a) Since the parity of the triangle is negative, the orientation is $U_{AB}^{(+,+)} \leftarrow U_{BC}^{(+,-)} \leftarrow U_{CA}^{(-,-)}$. (b) Two triangles sharing the edge $\{U_{AB}^{(+,+)}, U_{BC}^{(+,-)}\}$. (c) Four triangles sharing the edge $\{U_{AB}^{(+,+)}, U_{CA}^{(-,-)}\}$

This is an orientation because for every non-boundary edge, there are an even number of 2-simplices containing it, and they are paired by their opposite induced orientations. For example, consider the edge $\{U_{AB}^{\sigma_1}, U_{BC}^{\sigma_2}\}$, this edge only can be completed with a vertex indexed as $U_{CA}^{\sigma_3}$ for some compatible $\sigma_3 \in \{+,-\}^n$ constrained by the transitivity property. That is, for every component $i \in \{1, \dots n\}$, if $\sigma_1(i) = \sigma_2(i) = +$ (resp. $\sigma_1(i) = \sigma_2(i) = -$, then $\sigma_3(i) = +$ (resp. $\sigma_3(i) = +$). However, if $\sigma_1(i)$ and $\sigma_2(i)$ have different signs, both

signs are compatible in $\sigma_3(i)$. We can conclude that the admissible σ_3 are exactly 2^k (where k is equal to the number of voters i on the third situation). And, since by hypothesis $\{U_{AB}^{\sigma_1}, U_{BC}^{\sigma_2}\}$ is not in the boundary, $k > 0$.

Second, we can pair these 2^k 2-simplices saying that $\{U_{AB}^{\sigma_1}, U_{BC}^{\sigma_2}, U_{CA}^{\sigma_3}\}$ and $\{U_{AB}^{\sigma_1}, U_{BC}^{\sigma_2}, U_{CA}^{\sigma_3'}\}$ are paired if σ_3 and σ_3' are equal on each component but one. Then the parity associated to every triangle of a pair is opposite to the other member, so, their contribution on the edge $\{U_{AB}^{\sigma_1}, U_{BC}^{\sigma_2}\}$ determined by the induced orientations is also opposite. □

Now that we know that N_I is orientable we can use the index lemma to prove Theorem 2.2, for the case $|X| = 3, n = 2$.

Let $f : N_I \to N_O$ be a simplicial map such that for all vertices $U_{\alpha\beta}^{(+,\cdots,+)}$ of N_I, it holds that $f(U_{\alpha\beta}^{(+,\cdots,+)}) = U_{\alpha\beta}^{+}$. We use f to define a coloring of the vertices of N_I with colors $\{0, 1, 2\}$, and then use the index lemma (Theorem 3.1) to show that f is a projection.

In order to define the coloring of the vertices of N_I, first we colour them with $\{+1, -1\}$ according to the image of every vertex by f. That is, we label $U_{\alpha\beta}^{\sigma}$ with +1 iff $f(U_{\alpha\beta}^{\sigma}) \in N_O$ has the superindex +, and otherwise with -1. We call it the sign of $U_{\alpha\beta}^{\sigma}$ and it is denoted by $s(U_{\alpha\beta}^{\sigma})$.

Second, we color every vertex of N_I with one colour $p \in \{0, 1, 2\}$ following the rule:

$$p(U_{\alpha\beta}^{\sigma}) = ID(U_{\alpha\beta}^{\sigma}) + s(U_{\alpha\beta}^{\sigma}) \quad (mod\ 3) \tag{1}$$

where $ID(U_{AB}^{\sigma}) = 0$, $ID(U_{BC}^{\sigma}) = 1$ and $ID(U_{CA}^{\sigma}) = 2$ (for every $\sigma \in \{+, -\}^n$).

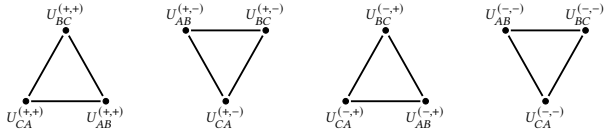


Figure 7: N_I has four boundary components generated by Condorcet cycles. A single triangle intersects each boundary edge since each pair of vertices determines the third one by transitivity.

Notice that a cycle of three vertices is 3-coloured if and only if the sign of all of them is the same. This implies that the content $C = 0$ because no 2-simplex in N_I can be mapped to one of the holes in N_O.

We conclude from the index lemma that $I = 0$, on the boundary of N_I, which consists of 4 combinations of Condorcet cycles (see Figure 7). The contribution to the index from the unanimity cycles is +2 (see Figure 8a).

Since the contribution of the unanimity cycles is +2 and $I = 0$, the two remaining contributions to I have to be -1 for each one of the remaining boundary components. So, we can conclude that both have to be tricoloured and mapped by f to the boundary of N_O.

Both of these boundary components cannot be mapped to the same boundary of N_O because if it were the case the simplex $\{U_{AB}^{(-,+)}, U_{BC}^{(-,+)}, U_{CA}^{(+,-)}\}$ would be mapped to one of the holes of N_O (see Figure 8b).

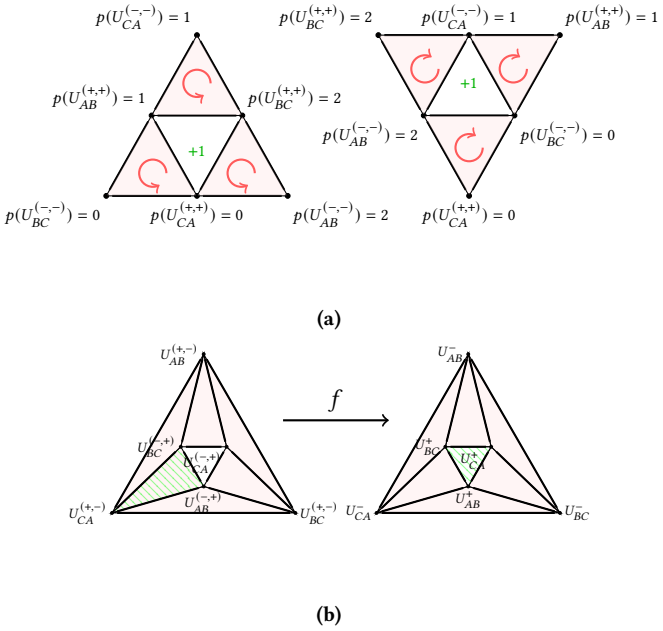


Figure 8: (a) The contribution of these two boundary components to the index is +2 according with the orientation exposed in Proposition 3.2 . (b) If these two boundary components of N_I are mapped to the same boundary component of N_O, then $U_{AB}^{(-,+)}$, $U_{BC}^{(-,+)}$ and $U_{CA}^{(+,-)}$ are also mapped to the same boundary component. That is, a hole.

Finally, we have all the information we need about the images of the 12 vertices of N_I to state that f is a projection. Recall that the images of the first and the fourth boundaries in Figure 7 are determined by the unanimity. If the second boundary is mapped to the inner boundary of N_O (and the third in the outer), it is straightforward to check that f is the projection over the first component. In contrast, if the second boundary is mapped to the outer boundary of N_O (and the third on the inner), then f is the projection over the second component.

4 APPLYING THE COMBINATORIAL TOPOLOGY APPROACH TO DOMAIN RESTRICTIONS

There is an extensive literature on the subject of domain restrictions, going back at least to Black [12] and Arrow [4], and the research area is still active today [8]. Arrow's impossibility applies to *universal domains*, where all possible individual preferences are considered. Many researchers have proved that it is possible to avoid Arrow's impossibility on some non-universal domains (see, e.g. [29, 41]). However, there is no general rule characterizing the domains in which aggregation is possible.

We illustrate here how the combinatorial topology approach can shed light on this topic. We provide a characterization of the

domain restrictions in which non-dictatorial aggregation is possible for $|X| = 3$ and $n = 2$. We also discuss the role of contractibility of the restricted domain, showing it is not what determines the possibility of avoiding Arrow's impossibility.

4.1 Arrow's impossibility using domain restrictions

We start with a domain restriction that exposes clearly a geometric reason for Arrow's impossibility, related to winding numbers. Namely, the following provides another proof of Theorem 2.2 for $|X| = 3$ and $n = 2$. It is the basis of the characterization of the domain restrictions in which non-dictatorial aggregation is possible.

A basic domain restriction. Recall the torus on the right of Figure 3. It consists of all the social profiles of N_I where the two voters disagree in either 1 or 2 of their pairwise preferences. The torus is depicted again in Figure 9, where in the top-left triangle, the profile is ABC, ACB, and there is disagreement in only one pairwise preference, BC, since the first voter prefers B over C and the second prefers C over B. In the following triangle on the left, the profile is BAC, ACB, with two pairwise disagreements, on BC and on AB. In fact, the torus is made of two triangulated cylinders, joined by the blue dotted circle and by the green dashed circle. The left cylinder is called C_1 and the right one is C_2. They are symmetric, if one exchanges the voter 1 and voter 2 in C_1 one gets C_2. Namely, the top-left triangle of C_1 is ABC, ACB, and the symmetric triangle in C_2 is ACB, ABC. Similarly for the next triangle of C_1, BAC, ACB, its symmetric triangle on C_2 is ACB, BAC.

Consider C_1 as a domain restriction of N_I, in Figure 9. It is obtained by removing the cylinder C_2 from the torus on the right of Figure 3, and removing also both of the concentric cylinders on the left of the figure, corresponding to unanimous profiles and those where the voters have opposite preferences. In Figure 9 all the triangles of C_2 are removed from the torus: from top to bottom, the triangles CAB, ABC, ACB, ABC, etc. Only the triangles on the left remain, which form the cylinder C_1. Notice that N_O is also a cylinder, except that the cylinder C_1 is subdivided into 12 triangles while N_O consists of 6 triangles. Denote by N_I' the resulting restricted domain.

Now, Arrow's geometric impossibility becomes clear: C_1 is wrapped once around N_O, and the wrapping is determined by the green-dashed cycle in C_1, due to unanimity. In the figure the image of the green-dashed cycle in C_1 on N_O is shown. This implies that the blue-dotted cycle, which is parallel to the green-dashed cycle, also has to wrap once around the cylinder, going in the same direction. There are two options for the aggregation function, labeled on the blue edges; to map the first (from top to bottom) blue edge to the edge 2 or to 5, the next one to 3 or 0 in N_O, and so on. In the first option the first voter is the dictator, in the second option the second voter is (in either case, the blue cycle goes on top of the green cycle of N_O).

Characterizing the maximal non-dictatorial restricted domains. Denote the cylinder on the left of the representation of the torus T as C_1 (see Figure 9) and the cylinder on the right as C_2. Assume that $f : N_I' \subseteq N_I \to N_O$ is an aggregation mapping satisfying unanimity. We have argued that if N_I' contains all of the triangles

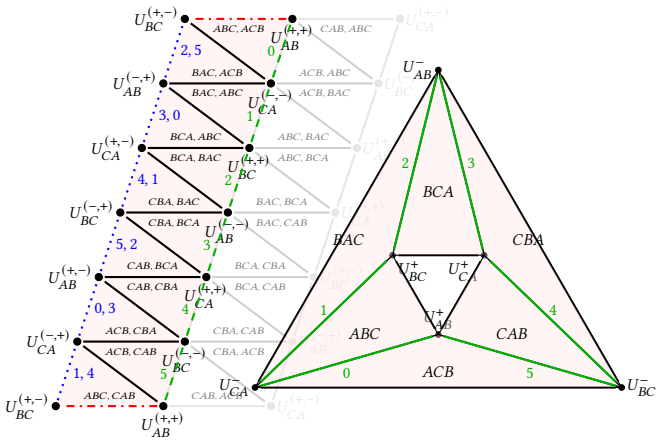


Figure 9: On the left is N_I', a domain restriction on N_I, resulting in a cylinder and how the green cycle is mapped to N_O. Inside of each triangle of N_I' is the corresponding individual preference; the top triangle is ABC, ACB, the next one BAC, ACB, and so on. The blue cycle has two labels on each of its edges; the first one is the social choice where the first voter is the dictator, from top to bottom, 2, 3, 4, 5, 0, 1. With the second labels, the second voter is the dictator.

containing a blue edge in one of the cylinders C_1 or C_2, then f is dictatorial.

Now, a simple analysis of what happens if we remove just one of the triangles with a blue edge in T, is used to derive the following.

Proposition 4.1. *Let $\mathcal{D}_T$ be the family of all subdomains of T admitting an unanimous and non-dictatorial aggregation function. The domains D satisfying the conditions below are maximal in $\mathcal{D}_T$ (with respect the inclusion):*

- *D is all T but two triangles, one in C_1 and the other in C_2.*
- *Both of these triangles contain a blue edge.*
- *Both blue edges must be different.*

Moreover, every domain contained in one of these maximal domains belongs to $\mathcal{D}_T$.

Finally, we can state the general characterization theorem, after proving an equivalent result to Proposition 4.1 but in N_I instead of in T.

Theorem 4.2. *Let $\mathcal{D}$ be the family of all subdomains of N_I, admitting an unanimous and non-dictatorial aggregation function. The domains D satisfying the conditions below are maximal in $\mathcal{D}$ (with respect the inclusion):*

- *D is all N_I but two triangles, one in C_1 and the other in C_2.*
- *Both of these triangles contain a blue edge.*
- *Both blue edges must be different.*

Moreover, every domain contained in one of these maximal domains belongs to $\mathcal{D}$.

Detailed proofs of these statements can be found on our preprint [49].

4.2 Eluding Arrow's impossibility while preserving non-contractibility

It has been argued that the existence of a rule that permits aggregation is related to contractibility of a topological space. For the existence case in the continuous setting (which is different from our Arrovian setting), Chichilnisky and Heal [20], and a 1954 topology theorem by Eckmann [23] show that, for a general class of domains, contractibility is necessary and sufficient. Building on this result and on Baryshnikov [10], for weak orders, Tanaka [51] shows a connection with Brower's fixed point theorem, in the case of $n = 2$ and $|X| = 3$. Baryshnikov [10] and other authors such as Lauwers [39] and Baigent [47] hypothesised in subsequent publications that the aggregation on non-universal domains could be equivalent to the contractibility of the induced input simplicial complex. That is, the aggregation á la Arrow on a domain $D \subseteq W^n$ would be possible iff the induced complex N_I' is contractible. Moreover, they added that in the well-known case of single-peaked preferences (in which aggregation is possible) contractibility is satisfied.

Next, we sketch a domain of preferences that proves that Baryshnikov's hypothesis above is not true. That is, we describe a domain N_I'' non-contractible and allowing non-dictatorial aggregation maps. For technical details and its graphical representation see our preprint [49].

The restricted domain N_I'' corresponds to a polarised society where political parties are classified as left-wing and right-wing parties. Assume that every left-wing voter will prefer all left-wing parties over all right-wing parties (vice-versa for right-wing voters). *A priori* we do not know if a voter is right-wing or left-wing.

We focus on the case in which there are two right-wing parties $\{A, B\}$ and one left-wing party C and two voters ($n = 2$). This way, N_I'' can be compared with the previous examples and proofs on this article.

The polarised domain restriction deletes the profiles in which a voter has C as the middle preferred party. For example, no voter will have the preference ACB because it prefer the right-wing party A over the left-wing party C and C over the right-wing party B.

There are non-dictatorial aggregation rules for N_I''. One of these rules is defined by two local dictators. The first voter is a local dictator between the right-wing parties A and B, whereas the second voter is a local dictator between a right-wing party and the left wing-party C. Formally, this aggregation map F is defined for every profile $\mathbf{R}$ in the domain as:

$$aF(\mathbf{R})b \Leftrightarrow aR_1b, \qquad aF(\mathbf{R})C \Leftrightarrow aR_2C, \qquad BF(\mathbf{R})C \Leftrightarrow BR_2C.$$

Using the fact that $aF(\mathbf{R})C \Leftrightarrow BF(\mathbf{R})C$, it is straightforward to check that F is well defined (i.e. $F(\mathbf{R})$ is transitive and complete for every $\mathbf{R}$). Additionally, F is unanimous, non-dictatorial and satisfies the independence of irrelevant alternatives.

It remains to check that N_I'' is not contractible. The proof is in our preprint[49].

5 REDUCTION TO THE CASE OF $n = 2$ AND $|X| = 3$

We have proved Arrow's impossibility Theorem 2.2 for $|X| = 3, n = 2$. The proof of Theorem 2.2 for $|X| \geq 3, n \geq 2$ follows directly from Lemma 5.1 and 5.2, given that the case $|X| = 3, n = 2$ has been proved.

Lemma 5.1. *Let the number of voters be any $n \geq 2$. Arrow's impossibility theorem for $|X| = 3$ implies it for $|X| \geq 3$.*

Proof. Suppose that Arrow's theorem is true when $|X| = 3$. We prove that for any X (with $|X| \geq 3$) and any $F : W^n \to W$ satisfying unanimity and independence of irrelevant alternatives, F is dictatorial.

Given F, choose three distinct alternatives $x, y, z \in X$ and denote $\overline{W}_0$ the set of all strict orders over these three alternatives. Define an aggregation map $\overline{F} : \overline{W}_0{}^n \to \overline{W}_0$ as follows. The image of a profile $(\overline{R}_1, \ldots, \overline{R}_n) \in \overline{W}_0{}^n$ by $\overline{F}$ is the restriction of the ordering $F(R_1, \ldots, R_n) \in W$ on the set $\{x, y, z\} \subseteq X$, where for each i, R_i is any extension of $\overline{R}_i$ from $\overline{W}_0$ to W. Notice that the definition of $\overline{F}$ does not depend on the chosen extension because of the independence of irrelevant alternatives of F. Moreover, it is easy to check that $\overline{F}$ satisfies unanimity as well as independence of irrelevant alternatives. So, it follows that $\overline{F}$ is dictatorial because we have supposed that Arrow's theorem is true when $|X| = 3$. It remains to prove that F is also dictatorial.

If k is the dictator of $\overline{F}$, we will prove that it is also a dictator for F. Consider a profile $\mathbf{R} = (R_1, \ldots R_n) \in W^n$ where aR_kb for some $a, b \in X$. Then take a profile $\mathbf{R}' = (R_1', \ldots R_n') \in W^n$ satisfying that, for every i, $xR_i'bR_i'aR_i'y$ if bR_ia, and $aR_i'yR_i'xR_i'b$ if aR_ib.

Since k is a dictator of $\overline{F}$ and $yR_k'x$ (k prefers a over b in R_k), we know that the image by $\overline{F}$ of the restriction of $\mathbf{R}'$ over $\overline{W}_0^n$ prefers y over x, hence $F(\mathbf{R}')$ also prefers y over x. Moreover, by unanimity, it holds that $aF(\mathbf{R}')y$ and $xF(\mathbf{R}')b$. Then, we obtain that $aF(\mathbf{R}')b$ from the relations $aF(\mathbf{R}')yF(\mathbf{R}')xF(\mathbf{R}')b$ using the transitivity. Finally, using the independence of irrelevant alternatives, we obtain that $aF(\mathbf{R})b$. Since this happens for every pair $a, b \in X$, k must be the dictator of F. □

The proof of the previous lemma, contrary to the ones in [2, 52], is not inductive. This fact enables us to reduce the cases of any cardinality of X to $|X| = 3$ in a single step.

Lemma 5.2. *Let the number of alternatives be any $|X| \geq 3$. If Arrow's impossibility theorem is true for $n = 2$ then it is true for $n > 2$.*

Proof. The proof is by induction on n. By hypothesis, the theorem is true when $n = 2$. Suppose that it is true for $n - 1$ and we will prove it for n.

Let $F^n : W^n \to W$ an aggregation map satisfying unanimity and independence of irrelevant alternatives. We will prove that F^n is dictatorial in three steps:

Step 1: We define the aggregation map on W^{n-1}, $F_1^{n-1}(R_1, \ldots, R_{n-1}) := F^n(R_1, \ldots, R_{n-1}, R_1)$. Since F_1^{n-1} satisfies unanimity and independence of irrelevant alternatives, the induction hypothesis guarantees that it has a dictator k_1. We will prove that if $k_1 \neq 1$, then k_1 is also a dictator for F^n.

Suppose $\mathbf{R} \in W^n$ and xR_ky. If the ordering of R_1 and R_n coincides on $\{x, y\}$, then $xF^n(\mathbf{R})y$ because F_1^{n-1} has k_1 as a dictator. Otherwise, we can suppose without loss of generality that xR_1y, yR_nx. Then, let $z \in X$ be an auxiliary alternative and let $\mathbf{R}' \in W^n$

be a profile which coincides with $\mathbf{R}$ over $\{x, y\}$, $xR'_{k_1}zR'_{k_1}y$ and z is below x and y for the remaining voters.

Since R'_1 and R'_n agrees on $\{y, z\}$ and k_1 is a dictator for F_1^{n-1}, we have that $zF^n(\mathbf{R}')y$. Moreover, $xF^n(\mathbf{R}')z$ because of the unanimity. Using the transitivity we obtain that $xF^n(\mathbf{R}')y$, and applying the independence of irrelevant alternatives we obtain that $xF^n(\mathbf{R})y$. So, k_1 is a dictator of F^n (if $k_1 \neq 1$).

Step 2: We define $F_2^{n-1}(R_1, \ldots, R_{n-1}) := F^n(R_1, \ldots, R_{n-1}, R_2)$. Using the inductive hypothesis, F_2^{n-1} has a dictator k_2. If $k_2 \neq 2$, apply a symmetric reasoning to the one in step 1 to deduce that k_2 is the dictator of F^n (if $k_2 \neq 2$).

Step 3: If $k_1 = 1$ and $k_2 = 2$, we show that n is the dictator of F^n. Let $\mathbf{R} \in W^n$ be a profile with xR_ny. Consider $z \in X$, and $\mathbf{R}' \in W^n$ coinciding with $\mathbf{R}$ over $\{x, y\}$, $xR'_nzR'_ny$, xR'_1z and zR'_2y. Using that 1 (resp. 2) is the dictator of F_1^{n-1} (resp. F_2^{n-1}) and the independence of irrelevant alternatives, we obtain that $xF^n(\mathbf{R})z$ (resp. $zF^n(\mathbf{R})y$). So, using transitivity, we obtain that $xF^n(\mathbf{R})y$. Finally we conclude that n is the dictator of F^n (if $k_1 = 1$ and $k_2 = 2$). □

The reader may wonder why Lemma 5.2 is inductive, instead of applying some direct argument extending from $n = 2$ to any number of voters (as we have done in Lemma 5.1). If such argument existed, it would allow to extend the theorem to an infinite number of voters. However, this is not possible because Arrow's impossibility is not true when n is infinite [27].

There are several works in which the proof of Arrow's theorem is only for $|X| = 3$ and/or $n = 2$ (e.g. [2, 21, 47, 51]). Using Lemma 5.1 and 5.2, all these proofs are extended to $|X| \geq 3$ and/or $n \geq 2$.

A few works have used inductive arguments over the number of voters or alternatives. In the fifties, Weldon [54] proved an impossibility theorem under a set of non-Arrovian axioms. Unlike our case, he could set the initial case of his inductive argument on the trivial case $n = 1$ (instead of $n = 2$). More recent works [2, 52] use inductive arguments using the base case $|X| = 3, n = 2$, as we do. However, our proof is more general. That is, whereas the results of Akashi [2, Lemma 1] and Tang and Lin [52, Lemma 1] are constrained to finite sets of alternatives, Lemma 5.1 works also for infinite X. In addition, the inductive step in [52, Lemma 2] is proved by contradiction using a large family of maps, while Lemma 5.2 uses only two, and using an explicit map that helps to understand the inductive step.

6 RELATION TO DISTRIBUTED COMPUTING

We start with a very brief overview of the notion of a task, and its representation using simplicial complexes, following the topology approach to distributed computing [35].

Distributed computability. A task is a distributed version of a function in sequential computing. When there is a single computing process, the function f specifies, for each possible initial input x, the value $f(x)$ that the process should compute. In a distributed system composed of several processes, each one gets only part of the input x. Thus, we may think of the input x as a simplex $\{(p_1, x_1), (p_2, x_2), \ldots, (p_n, x_n)\}$, for n processes, specifying that the private input of p_i is x_i. Then, the processes run their individual programs, communicating with each other (through message passing or a shared memory), and eventually produce individual local output values, defining a simplex $\{(p_1, y_1), (p_2, y_2), \ldots, (p_n, y_n)\}$, where y_i is the output value of process p_i. A task $\langle \mathcal{I}, \mathcal{O}, \Delta \rangle$ consists of an input complex $\mathcal{I}$, of all valid input simplexes, an output complex $\mathcal{O}$ of all valid output simplexes, and by an input/output relation Δ, that specifies, for each possible input simplex x, a set $\Delta(x) \subset \mathcal{O}$ of valid output simplexes.

The most important task is *consensus*. In its binary form, the possible inputs x_i are taken from the set $\{0, 1\}$, and there are only two possible output simplexes: either everybody decides 0 or everybody decides 1. Then, $\Delta(x)$ states that if everybody starts with the same input, then everybody decides that input, else, it is valid to decide either of the two output simplexes.

Thus, each input simplex $\sigma \in \mathcal{I}$ defines an initial configuration of the distributed system. After the processes run their local algorithms and communicate with each other, they eventually stop, and end up in a final configuration τ. The simplex τ is of the same form of the input and output simplexes, except that in a vertex (P_i, x_i) of τ, x_i denotes the final local state of process P_i. An algorithm solving the task, defines a *decision function* δ. This local state x_i determines the output value decided by the process P_i, which is $\delta(id_i, x_i)$.

Notice that there may be many possible runs all starting on input σ, because of possible failures, different speed of execution of the processes, etc. The set of all possible final configurations can also be represented as a chromatic complex, denoted $\mathcal{P}(\sigma)$. The *protocol complex*, $\mathcal{P}$, is the union of $\mathcal{P}(\sigma)$, over all $\sigma \in \mathcal{I}$. The task is solved if and only if there exists a chromatic simplicial map δ from $\mathcal{P}$ to $\mathcal{O}$ respecting Δ, such that $\delta(\mathcal{P}(\sigma))$ is contained in $\Delta(\sigma)$. The simplicial map δ is chromatic in the sense that it sends vertices to vertices preserving process ids.

Fault-tolerant elections. Consensus is similar to an election, where each process votes for a single alternative, with no further requirements, except the validity requirement that the elected alternative has to be voted by at least one process. If processes can communicate with each other and learn their respective inputs (votes), then they can apply a function F to the input simplex σ and solve consensus, where $F(\sigma)$ would be the social preference. Consensus is known to be unsolvable by n asynchronous processes even if only one of them can fail by crashing [26], but in synchronous systems it is solvable, even if processes can fail in arbitrary, malicious Byzantine ways [48], as long as the number of failed processes f is $f < n/3$. The consensus problem when the inputs of the processes (votes) are strict total orderings of the alternatives has also been considered [18]. Several validity conditions are studied, that impose requirement on the ranking decided based upon the preferences provided by the processes that do not fail, and possibilities and impossibilities are shown, depending on the relation between f and n. For subsequent work see [43, 53].

Arrow's impossibility and distributed computing. Thus, the question of existence of a non-dictatorial aggregation function F in a distributed system can be understood in a setting where processes can agree on their inputs, and then each one applies F locally; similar to the setting of [18] where a Byzantine agreement protocol is used to agree on the inputs of the processes. What we show (Arrow's impossibility) is that the problem cannot be solved, even in a failure-free setting, if F is required to satisfy unanimity and independence of irrelevant alternatives.

The intuition behind our impossibility proof, for the base case of two voters and three alternatives A, B, C is the following. Recall the schematic illustration of N_I and N_O in Figure 1. The input complex is N_I while the output complex is N_O. The processes of the task correspond to the pairs of alternatives $\mathcal{P} = \{AB, BC, AC\}$, called also *ids*. Thus, we consider *chromatic* simplicial complexes, where the vertices of each triangle are labeled with distinct process ids from $\mathcal{P}$. There are four possible individual inputs $\{++, --, +-, -+\}$, while the possible individual outputs are $\{+, -\}$. The output complex N_O consists of all chromatic triangles, with each vertex labeled with an output value from $\{+, -\}$, except for the two triangles labeled with the same value. Thus, N_O is the output complex of the *weak symmetry breaking* task e.g. [16, 38, 42]. Similarly, N_I consists of all chromatic triangles whose vertices are labeled with input values from $\{++, --, +-, -+\}$, except the 28 triangles whose vertices have the same sign in the first or in the second component. Thus, N_I includes the (torus) complex of the renaming task [17], where every triangle is labeled with distinct values from $\{++, --, +-, -+\}$, plus 12 additional triangles where one value repeats twice.

The second step is to observe that the aggregation map F, that decides the social output, induces a simplicial map f from N_I to N_O, which is chromatic (preserves vertex ids). Notice that while in the input complex $\mathcal{I}$ of a task the indistinguishability [6] comes from the local nature of the inputs, in N_I it is a consequence of the requirement of independence of irrelevant alternatives.

The topological reformulation Arrow's theorem (Section 2.3) is analogous to a task specification of distributed computing: we seek a chromatic simplicial map f from N_I to N_O, that sends vertices with input ++ to vertices with output + and vertices with input $--$ to vertices with output $-$. This comes from the requirement that if both voters prefer an alternative x over y, then the social preference should prefer x over y. Arrow's impossibility reformulation Theorem 2.2 says that such an aggregation map f must be a dictatorship.

We get a task specification on a restricted domain (of all possible quaternary input values). Similarly to social choice, where first Arrow's impossibility was proved, and then restricted domains to avoid it have been studied, in distributed computing first the consensus impossibility was proved, and then restricted domains (called *conditions*) have been studied in a series of papers since [45].

Interestingly, the index lemma is also behind the distributed computing impossibilities related to weak symmetry breaking e.g. [16, 33, 38, 42].

7 CONCLUSIONS

We have given new proofs of Arrow's theorem consisting of two parts. The first part deals with the base case of two voters and three alternatives, and we presented three different versions: using the index lemma, using pivotal voters (in the full version [49]), and using domain restrictions. The second part proves the general case by a simple reduction to the base case.

The first part shows that any aggregation function is dictatorial, because in essence it is mapping a torus onto a cylinder, in a continuous way, respecting unanimity. The argument sheds light on the algebraic topology proof of Baryshnikov [10], and makes it accessible to a wider audience. Furthermore, it connects it to standard proofs of Arrow's theorem based on pivotal arguments, by explaining how the paths of such arguments move along the torus and the cylinder [49].

The structure of our proofs, in two parts, suggests that the interesting structure happens in the base case, $|X| = 3, n = 2$. We have considered domain restrictions on the base case, showing that there is a domain restriction where Arrow's impossibility is derived from the geometry in an intuitive way, and there is another domain restriction where it does not hold, yet it is not contractible. Furthermore, a characterization of the domain restrictions that allow non-dictatorial aggregation can be derived using this type of ideas. We leave the generalization of the characterization to $|X| \geq 3, n \geq 2$ for future work.

We hope that bringing in combinatorial topology to social choice problems opens interesting opportunities for future work. These tools have been encountering many applications recently. Some examples are in concurrency [1], image processing [7], political structures [44], data analysis [37] and wireless networks [50].

Combinatorial topology has been very useful in distributed computing [35]. We described some analogies that are worth exploring further, since computing processes that communicate with each other need to agree on one of their inputs in many applications. Remarkably, while Sperner's lemma is the key to the impossibilities of tasks where processes need to reach agreement such as *consensus*, *set agreement* [5], *vector consensus* [46] and *interactive consistency* [28] (where domain restrictions are studied), for Arrow's impossibility, the key is the index lemma, as it is for tasks related to renaming and weak symmetry breaking [16, 33]. Notice that renaming is weaker than set agreement and related tasks [15, 31].

Here we studied only the case where the aggregation map F is computed by a central entity that knows the votes of all participants. It would be interesting to explore the case where the agents can communicate with each other and subdivisions of the input complex arise, as in [18, 43, 53], where distributed applications where ranking voting is important are discussed, such as machine learning. Interestingly, the complexes N_I, N_O can no longer be used to study voting methods that do not satisfy the independence of irrelevant alternatives axiom, such as Condorcet methods, and in particular the Kemeny method used in [43]. Notice that the index lemma is preserved under subdivisions e.g. [33, Corollary 4]. However, we are not aware of a distributed task where the impossibility is proved in dimension 2, and then extended easily to any dimension.

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